KASIN ITO-SATO

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Mechanical Engineer (BS, May 2027) | Mechanical Design | Testing & Analysis | HVAC/MEP | GD&T

EDUCATION

Loyola Marymount University **May 2027**

Bachelor of Science (BS), Mechanical Engineering

Relevant Coursework: Machine Design, Mechanics of Materials, Heat Transfer, Fluids, Thermodynamics, Dynamics & Control, Propulsion (grad)

PROFESSIONAL EXPERIENCE

WSP in the US **May 2025 - January 2026**

Mechanical Engineering Intern *Honolulu, HI*

- **3 Excel calculation tools adopted team-wide** for building pressurization, demand-controlled ventilation, and exhaust rates, built on my own initiative and documented in walkthrough guides citing ASHRAE 90.1 and 62.1.
- **Showed the team's 10% rule of thumb left buildings under-pressurized** (0.0026 vs. 0.02 in. w.c. minimum on a 40,000 sq ft office) using my pressurization tool.
- **Sized HVAC equipment from loads cross-checked in 2 models** (IES-VE, TRACE 700), calculated duct pressures, and reviewed manufacturer submittals against specifications.
- **Drafted HVAC and plumbing layouts in Revit and AutoCAD** for the Hawaii Prince Waikiki Spa Renovation through multiple design stages as the owner and architect issued changes.
- **Caught a plan vs. installation discrepancy** (storm-drain junction) during a site visit to the Ulana High-Rise; confirmed it was an intentional field change for space.

ENGINEERING PROJECTS

Rope-Climbing Robot | Machine Design, LMU Spring 2026

- **Crossed a 38 ft outdoor rope span in 5.5 minutes** (0.115 ft/s, 86% of the 0.134 ft/s kinematic prediction) with a robot whose 3-stage spur gear drivetrain (64T/32T/16T) I built and tested on a four-person team.
- **Pivoted from the matrix-selected arm design to a constrained screw slider** after lab testing showed the arms would not work, iterating hooks, slider, and gear engagement with wood and 3D-printed parts.

Structural & Materials Testing | Mechanics and Materials Science Labs, LMU 2025

- **Generated S-N curves for 7075-T6 aluminum and 4140 steel** from rotating-beam fatigue tests at 6,000 rpm (26 to 80 ksi), identifying an endurance limit in steel that aluminum lacked.
- **Measured buckling stress in 6061-T6 aluminum tubes, which fell from 35.5 to 11.4 ksi** as length rose from 4.5 to 10 in, and compared aluminum and steel results to Euler theory.
- **Measured 67.5 ksi yield and 77.5 ksi ultimate strength** in 7050-T7 aluminum through ASTM E8 tensile tests on an Instron 4505, within about 2% of published yield.

Heat Sink Testing & Snow-Melt System Design | Heat Transfer, LMU Spring 2026

- **Cut block temperature from 137.5 to 89.0°C with a fin array** and measured convection coefficients of 12.8 W/m²K (free) and 48.3 W/m²K (forced) using Arduino thermocouple data acquisition.
- **Sized hydronic piping for a 1,500 sq ft snow-melt driveway** by modeling convection, soil conduction, and pipe pressure losses in MATLAB, finding the smallest pipe (NPS 2) that met a 0.8 in/hr melt target.

SAE Aero Design (Regular Class 2027) | Capstone Design, LMU 2026 - 2027

- **Derived requirements from 2027 SAE rules** (55 lb, 120 in, 2200 mAh) for a five-person team, traded two- vs. four-motor configurations, and laid out motor, ESC, and battery integration.

2.0x Pantograph | Design and Manufacturing, LMU Fall 2024

- **1.98x to 2.00x measured accuracy** on an aluminum linkage we machined from drawings I helped detail with true-position GD&T at MMC (ASME Y14.5).

SKILLS

CAD & Drawings: SolidWorks (parts, assemblies, drawings), Onshape, AutoCAD, Revit, Rhinoceros, GD&T (ASME Y14.5)
Analysis & Programming: Hand calculations, MATLAB/Simulink, root locus control design, Arduino C++, CES EduPack, Excel
Test & Fabrication: Tensile, fatigue, buckling testing; thermocouple DAQ; mill, lathe, CNC, laser cut, waterjet, 3D print, welding
HVAC/MEP: Loads (IES-VE, TRACE 700), equipment selection, duct pressure, ASHRAE 62.1/90.1, IECC, UPC, submittals
Certifications: FE Mechanical exam scheduled November 2026; Rhinoceros CAD Level 2 **Languages:** English, Japanese